# Rational differences of periodic Dyck-path generating functions

Principia Math In Progress

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## Task 1: Complete solution of MathDB 320607

Eu and Fu's Conjecture 5.2 [1] gives two identities relating generating functions for up-step heights modulo $m$. We prove both as special cases of a single identity. Here $m$ is a height modulus for ordinary Dyck paths, as defined in the source, not the arity of a path. The symbol $t$ in the source's factors $1-t$ denotes the marking variable $y$ used everywhere else in those formulas. We prove the intended identities with $1-y$; no credit is claimed for a notational correction.

An up step ending at height $h$ has height $h$. For a Dyck path $\pi$ of semilength $n$, let $a_c(\pi)$ count its up steps with $h \equiv c \pmod m$. Define

$$G^{(c;m)} = G^{(c;m)}(x,y) = \sum_{n\geq 0} \sum_{\pi\in\mathcal{D}_n} x^n y^{a_c(\pi)}.$$

All identities below are in the formal series ring $\mathbb{Q}[y][[x]]$. Let the continuant polynomials be

$$D_{-1} = 0, \qquad D_0 = D_1 = 1, \qquad D_n = D_{n-1} - xD_{n-2} \quad (n \geq 2).$$

The recurrence also holds at $n = 1$ with these conventions.

**Theorem 1.** *For integers $i \geq 1$ and $m \geq 2i+2$,*

$$G^{(m-i-1;m)} - G^{(i;m)} = \frac{(1-y)x^i D_{m-2i-2}}{D_{m-1} + x(1-y)D_{i-1}D_{m-i-2}}. \tag{1}$$

*In particular, writing $u = 1/(2\sqrt{x})$, the following two identities hold:*

$$G^{(m-2;m)} - G^{(1;m)} = \frac{(1-y)U_{m-4}(u)}{U_{m-2}(u) - y\sqrt{x}\,U_{m-3}(u)}, \qquad m \geq 4, \tag{2}$$

$$G^{(m-3;m)} - G^{(2;m)} = \frac{(1-y)U_{m-6}(u)}{U_{m-2}(u) - yU_{m-4}(u) + \sqrt{x}\,U_{m-5}(u)}, \qquad m \geq 6. \tag{3}$$

*These are both parts of Conjecture 5.2.*

### Periodic continued fractions

Give an up step at height $h$ weight $xy$ when $h \equiv c \pmod m$, and weight $x$ otherwise. A first-return decomposition $U\mu D\nu$ gives the continued fraction with successive numerators equal to those weights. Equivalently, for $1 \leq c \leq m-2$, put

$$M = \begin{pmatrix} 0 & 1 \\ -x & 1 \end{pmatrix}, \qquad M_y = \begin{pmatrix} 0 & 1 \\ -xy & 1 \end{pmatrix}, \qquad P_c = M^{c-1}M_yM^{m-c} = \begin{pmatrix} A_c & B_c \\ C_c & E_c \end{pmatrix}.$$

A matrix acts by $z \mapsto (az+b)/(cz+d)$, so $M$ and $M_y$ act by $1/(1-xz)$ and $1/(1-xyz)$, respectively. One full period starts at height one, with the marked position at $c$. Thus the source's established continued fraction, Theorem 3.1, gives

$$C_c(G^{(c;m)})^2 + (E_c - A_c)G^{(c;m)} - B_c = 0. \tag{4}$$

This also follows directly from the first-return decomposition just given. The combinatorial series has constant coefficient one.

### Cancellation of the common quadratic term

Cyclic invariance of trace and multiplicativity of determinant imply

$$\operatorname{tr} P_c = \operatorname{tr}(M_y M^{m-1}) =: T, \qquad \det P_c = x^m y,$$

independently of $c$. From (4), the series

$$S_c = 2C_c G^{(c;m)} + E_c - A_c$$

satisfies $S_c^2 = T^2 - 4x^m y$. At $x = 0$, all matrices $P_c$ equal $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, hence $S_c(0, y) = 1$. For any $c, d$, the identity $(S_c - S_d)(S_c + S_d) = 0$, and the invertibility of $S_c + S_d$, show that

$$S_c = S_d. \tag{5}$$

This argument fixes the formal square-root branch without an analytic choice.

Induction on $n$ gives, for $n \geq 1$,

$$M^n = \begin{pmatrix} -xD_{n-2} & D_{n-1} \\ -xD_{n-1} & D_n \end{pmatrix}.$$

Since $M_y - M = x(1-y)\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, matrix multiplication yields

$$A_c = -xD_{m-2} - x^2(1-y)D_{c-2}D_{m-c-2}, \tag{6}$$

$$C_c = -x\big(D_{m-1} + x(1-y)D_{c-1}D_{m-c-2}\big). \tag{7}$$

For $c = 1$, the left factor is the identity matrix; the same formulas hold because $D_{-1} = 0$ and $D_0 = 1$.

Now set $c = i$ and $d = m - i - 1$. Formula (7) gives $C_c = C_d =: C$. Since the traces are equal, $E_c = T - A_c$ and $E_d = T - A_d$. Subtracting (5) therefore gives

$$C(G^{(d;m)} - G^{(c;m)}) = A_d - A_c. \tag{8}$$

The elementary continuant identity

$$D_a D_b - D_{a-1}D_{b+1} = x^a D_{b-a} \qquad (0 \leq a \leq b) \tag{9}$$

follows by reducing $a$ by one using the recurrence; the case $a = 0$ is immediate. Applying it with $a = i - 1$ and $b = m - i - 3$ in (6) gives

$$A_d - A_c = -x^{i+1}(1-y)D_{m-2i-2}.$$

Substitution in (8) and cancellation of $-x$ prove (1). Its denominator has constant coefficient one, so the division is valid in the stated formal series ring.

### The two conjectured formulas

The Chebyshev recurrence gives

$$D_n = x^{n/2} U_n(1/(2\sqrt{x})) \quad (n \geq 0).$$

This expression is a polynomial despite its notation. At $i = 1$, the denominator of (1) reduces to

$$D_{m-1} + x(1-y)D_{m-3} = D_{m-2} - xyD_{m-3}.$$

At $i = 2$, it reduces to

$$D_{m-1} + x(1-y)D_{m-4} = D_{m-2} - xyD_{m-4} + x^2 D_{m-5}.$$

Substituting the Chebyshev expression and canceling the common powers of $x$ gives (2) and (3), respectively. All endpoint indices are nonnegative, including $m = 4$ and $m = 6$. This proves the theorem.

The verifier uses exact arithmetic to compare the formula with direct Dyck-path enumeration, longer first-return recurrences, matrix invariants, and the Chebyshev specializations. The finite checks support the proof; they do not supply its all-index conclusion. Review is by the authoring agent only, with no independent peer review or proof-assistant certification.

## Task 2: A bijective explanation

Find a path involution or cancellation explaining (1), with an explicit description of the paths left over. The source's involution for residues 0 and $m - 1$ suggests looking for a wider family of strip reflections, but the present proof is algebraic.

## Task 3: Several marked residues

Replace the single marked residue by a subset of residues, possibly with separate variables. Classify pairs of cyclic shifts for which the relevant transfer matrices have equal lower-left entries, and determine which resulting rational differences have combinatorial interpretations.

## Task 4: Coefficient asymptotics

For fixed $m, i$ and positive $y$, analyze the poles of the rational difference in (1). Determine how quickly the corresponding weighted counting sequences approach each other relative to their common dominant growth.

## Task 5: Grand down steps

For genuine $s$-ary paths with down steps $(1, -s)$ and $s > 1$, determine which height-residue statistics admit analogous rational differences. This is a separate extension; the theorem above concerns ordinary Dyck paths. Tasks 2–5 are questions, not additional solved MathDB problems.

## References

[1] S.-P. Eu and T.-S. Fu, *Exterior Pairs and Up Step Statistics on Dyck Paths*, Electronic Journal of Combinatorics **18**(1) (2011), P92, 15 pp., published article; arXiv:1010.5673v1.